Adaptive Traffic Signal Control Using Demand Forecasting and Rolling-Horizon Optimization

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***Abstract*— Traffic signal control plays an important role in managing congestion and reducing vehicle delays at urban intersections. Fixed-time controllers rely on predetermined timing plans, and queue-responsive controllers react to observed queues, but neither uses recurring temporal patterns in future arrivals. This project proposes a forecast-informed signal controller that integrates Transformer-based demand forecasting with rolling-horizon optimization. In our isolated-intersection simulation, vehicles follow the Intelligent Driver Model (IDM), so queue discharge, queue storage, and clearance needs result from car-following behavior rather than being assumed. We will compare the proposed controller with fixed-time and queue-responsive control under balanced demand, a directional demand shift, detector failure, and heavy demand. By examining the optimizer's internal queue model, we argue that forecasts can influence signal decisions only when a queue is expected to clear within the planning horizon, so the benefit of forecasting should depend on how congested the intersection is. All codes used in this project is available at `https://github.com/Kimyounggun99/CEE281-term-project.git`.**

I. CORE CONCEPTS

Our system environment and our approaches are summarized in Fig. 1. Our controller combines three ideas: how a signalized approach serves vehicles, how individual vehicles move and form queues, and how forecasts can be used inside an optimization. This section explains each idea and its assumptions. Units are seconds (s), meters (m), and vehicles (veh) unless stated otherwise.

A. Flow, Headway, and Intersection Capacity

Traffic flow q [veh/s] is the number of vehicles passing a point per unit time, and density k [veh/m] is the number of vehicles per unit length. They are linked by the mean speed $\bar{v}$ [m/s] through $q = k\bar{v}$,

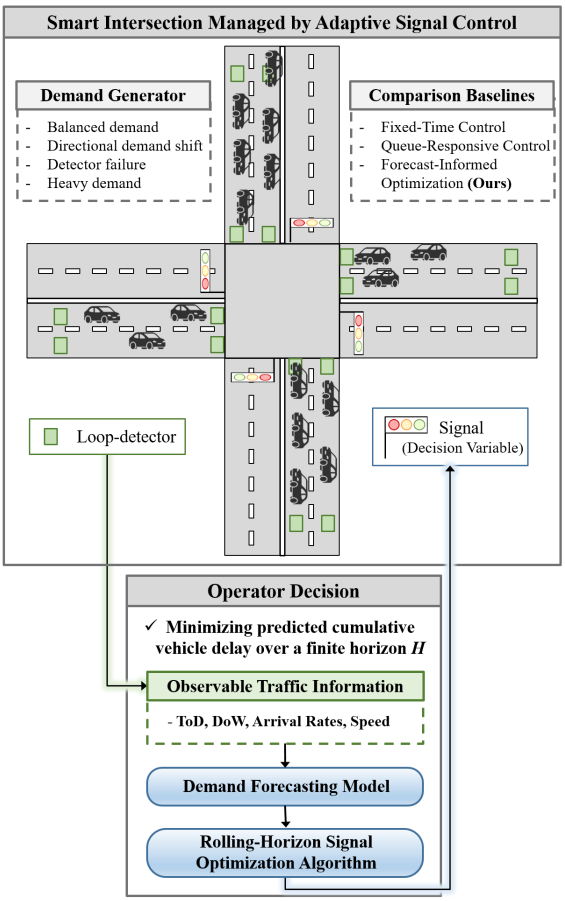


Fig. 1. Overview of the proposed adaptive traffic signal control framework integrating Transformer-based demand forecasting and rolling-horizon optimization in an isolated intersection.

and the average time headway between vehicles is $1/q$ [1].

When a signal turns green, the vehicles in a queue cross the stop line one after another. After the first few vehicles, which need extra time to react and accelerate, the queue discharges with a nearly constant headway h_s. The rate $S = 1/h_s$ [veh/s/lane] is the saturation flow. It is the key quantity in signal timing because it tells us how

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many vehicles a given green time can serve.

Let $\mathcal{A}$ denote the set of approaches. For approach $i \in \mathcal{A}$ with N_i lanes, arrival rate λ_i [veh/s], and effective green time g_i in a cycle of length C, the capacity c_i, flow ratio y_i, and degree of saturation x_i are

$$c_i = \frac{N_i S\, g_i}{C}, \qquad y_i = \frac{\lambda_i}{N_i S}, \qquad x_i = \frac{\lambda_i}{c_i}. \quad (1)$$

The flow ratio is the fraction of the cycle that approach i would need to be green. Each phase change also wastes time (yellow, all-red, and start-up), which we call the lost time L per cycle. All demand can therefore be served only if the sum of critical flow ratios satisfies $Y = \sum_i y_i < 1 - L/C$.

Webster [2] derived the cycle length that approximately minimizes delay, and the average delay caused by waiting through red:

$$C_0 = \frac{1.5L + 5}{1 - Y}, \qquad d_{u,i} = \frac{C(1 - g_i/C)^2}{2(1 - y_i)}. \quad (2)$$

The author also showed that random arrivals add delay that grows without bound as x_i approaches one. Intuitively, when an approach has spare capacity, a poor green allocation in one cycle is absorbed in the next. Near capacity there is no spare green to recover, so the same mistake leaves vehicles waiting for several cycles. This is why information about future arrivals should matter most when the intersection is busy but not yet overloaded.

B. Queue Formation and Discharge

Queues are conveniently described with cumulative counts. Let $A_i(t)$ be the number of vehicles that have arrived at approach i by time t, and $D_i(t)$ the number that have departed. The number waiting is $Q_i(t) = A_i(t) - D_i(t)$, and the area between the two curves, $\int Q_i\, dt$ [veh·s], equals total delay [3]. During green, D_i grows at most at rate $N_i S$; during red it stays flat. If arrivals exceed capacity for several cycles, the queue left after each green grows by $\lambda_i - c_i$ per second, and re-timing alone cannot remove it.

Real queues also occupy space. Each stopped vehicle needs its length l plus a standstill gap s_0, so an approach of length ℓ_a stores at most $\lfloor \ell_a/(l+s_0) \rfloor$ vehicles per lane. During red, the back of the queue moves upstream as new vehicles stop; after the light turns green, a start-up wave travels backward through the queue as each driver begins to move. Both boundaries move at the kinematic-wave speed [4]

$$w = \frac{q_u - q_d}{k_u - k_d}, \quad (3)$$

where (q_u, k_u) and (q_d, k_d) are the traffic states on either side of the boundary. When the queue tail reaches the upstream end of the approach, arriving vehicles can no longer enter, and the queue spills back beyond the modeled road. How quickly drivers react and accelerate inside the queue, and therefore how fast it discharges, depends on the car-following behavior described next [1].

C. Vehicle-Motion Model: the Intelligent Driver Model

The IDM [5] describes each driver as balancing two goals: reaching a desired speed and keeping a safe gap behind the vehicle ahead. We use it because its parameters have clear physical meanings, it avoids collisions under normal settings, and one equation reproduces free driving, following, stopping at a red light, and accelerating away from a queue.

Vehicle n has front-bumper position x_n [m] and speed v_n [m/s], with $\dot{x}_n = v_n$. The driver observes the net gap to the leader $s_n = x_{n-1} - x_n - l$ [m] and the approach rate $\Delta v_n = v_n - v_{n-1}$ [m/s], which is positive when closing in. The acceleration is

$$\dot{v}_n = a \left[1 - \left(\frac{v_n}{v_0} \right)^{\delta} - \left(\frac{s^*(v_n, \Delta v_n)}{s_n} \right)^2 \right], \quad (4)$$

with the desired gap

$$s^*(v, \Delta v) = s_0 + \max\!\left(0,\; vT + \frac{v\, \Delta v}{2\sqrt{ab}} \right). \quad (5)$$

Here v_0 is the desired speed [m/s], T the desired time gap [s], s_0 the standstill gap [m], a the maximum acceleration [m/s^2], b the comfortable deceleration [m/s^2], and δ [–] controls how acceleration fades near v_0.

The bracket in (4) contains two competing effects. The term $(v_n/v_0)^{\delta}$ lets the vehicle accelerate on an empty road and stop accelerating near v_0. The term $(s^*/s_n)^2$ brakes the vehicle when its gap is shorter than desired. The desired gap grows with speed through vT, so faster drivers keep more

distance, and it grows with the closing speed Δv, so a driver approaching a stopped vehicle brakes early and smoothly. When the leader pulls away ($\Delta v < 0$), the driver accelerates.

If all vehicles travel at the same constant speed, $\dot{v}_n = \Delta v_n = 0$, and (4) gives the equilibrium gap and flow

$$s_e(v) = \frac{s_0 + vT}{\sqrt{1 - (v/v_0)^{\delta}}}, \qquad Q_e(v) = \frac{v}{s_e(v) + l}. \tag{6}$$

The maximum of Q_e is the capacity the model can sustain. An important consequence for our project is that saturation flow is not a separate input: it follows from how drivers accelerate and how much gap they keep, and it is usually below $\max Q_e$ because vehicles leaving a queue start from rest.

Interaction with signals. A red light acts as a stopped virtual vehicle at the stop line ($x = 0$): the driver uses $s_n = -x_n$ and $v_{n-1} = 0$ when this is closer than the real leader, and so stops smoothly. At the start of yellow, a driver too close to stop comfortably (distance below $v_n^2/2b$) continues through. Using the kinematic rules of the Signal Timing Manual [6], a 1-s reaction time, 3 m/s^2 deceleration, and 10 m/s speed give a yellow of about $1 + 10/6 \approx 2.7$ s (we use 3 s), and clearing a 24-m intersection takes about $(24 + 4.5)/10 \approx 2.9$ s, which indicates how long the all-red must last.

Parameters and numerics. Table I compares our parameters with published values. Published calibrations use $s_0 = 2$ m, $\delta = 4$, and b of 1.67–2 m/s^2 [5], [7]. The model author describes T of 0.8–2 s and a of 0.8–2.5 m/s^2 as realistic and recommends a lower v_0 for city traffic [8]; we use 10 m/s (36 km/h). By (6), these parameters give a capacity of about 1340 veh/h/lane. We update speed and position every 0.2 s assuming constant acceleration over the step, which gives essentially the same results for any step below 0.5 s [8]. Braking is limited to 6 m/s^2, and any position correction needed to prevent overlap is counted.

D. Demand Forecasting and Rolling-Horizon Control

Traffic arrivals often follow recurring patterns associated with the time of day and the day of the week, and forecasting methods can capture these patterns in intersection volumes [9]. A Transformer [10] is a neural network that predicts the next value of a sequence by letting each past time step attend to the others, so it learns which earlier intervals are most informative for the prediction. Because attention by itself ignores order, a positional encoding tells the model when each input occurred. To forecast several steps ahead, each prediction is fed back as an input; this is simple, but errors can accumulate as the horizon grows.

A rolling-horizon controller plans the signal sequence over a short future horizon using predicted arrivals, applies only the first decision, and plans again when new detector data arrive. Because the plan is refreshed frequently, measurements can correct part of a wrong forecast [11], [12]. Other approaches treat the uncertainty in arrivals explicitly [13] or act only on currently observed queues, as in max-pressure control [14]; our queue-responsive baseline follows the latter idea.

TABLE I

IDM PARAMETERS USED AND LITERATURE VALUES

Symbol [unit]	This study	Literature
v_0 [m/s]	10	33.3 (highway) [5], [7]
T [s]	1.5	1.5 [7]; 0.8–2 [8]
s_0 [m]	2	2 [5], [7]
a [m/s^2]	1.2	0.73 [5]; 1.4 [7]
b [m/s^2]	2.0	1.67 [5]; 2.0 [7]
δ [–]	4	4 [5]
l [m]	4.5	–
Δt [s]	0.2	< 0.5 [8]

II. HOW WE WILL BUILD UPON THEM

A. Research Question and Prediction

Question. Under balanced demand, a directional demand shift, detector failure, and heavy demand, when does forecast-informed rolling-horizon optimization reduce total intersection delay compared with fixed-time and queue-responsive control, and how do forecast errors affect that benefit?

Why the answer should depend on the condition. Consider an intersection where both directions have long queues that will not clear within the next minute. Every vehicle arriving in that minute will wait no matter which phase is green, so the arrivals add the same delay to every candidate signal plan, and the forecast cannot help the controller choose

between them. The forecast becomes useful only when the controller expects a queue to empty: then it must decide whether to keep the green for vehicles that are about to arrive or to switch.

The optimizer's internal model makes this precise. Following our original formulation, the controller predicts the queue on approach i second by second as

$$\hat{q}_{i,\tau+1} = \max\{0,\ \hat{q}_{i,\tau} + \hat{\lambda}_{i,\tau}\Delta t - m_{i,\tau}\hat{S}_i\Delta t\}, \quad (7)$$

where $\hat{\lambda}_{i,\tau}$ is the Transformer's forecast arrival rate, $m_{i,\tau} \in \{0, 1\}$ indicates green, $\hat{S}_i$ [veh/s] is the assumed discharge rate, and $\Delta t = 1$ s. It selects the signal decisions $\mathbf{u}$ that minimize the predicted cumulative delay

$$J(\mathbf{u}) = \sum_{\tau=t}^{t+H-1} \sum_{i\in\mathcal{A}} \hat{q}_{i,\tau}\Delta t. \quad (8)$$

If no predicted queue reaches zero within the horizon H, the $\max$ never activates, and J separates into

$$J(\mathbf{u}) = J_0\big(\hat{\mathbf{q}}_t, \hat{\boldsymbol{\lambda}}\big) - \sum_{\tau,i} \hat{S}_i\, G_i(\mathbf{u}; t, \tau)\, \Delta t, \quad (9)$$

where J_0 contains the current queues and forecast arrivals but not $\mathbf{u}$, and G_i is the cumulative green time given to approach i. The forecast thus drops out of the decision. The remaining term rewards giving as much total green as possible, and since every phase change costs 4 s of yellow and all-red, the controller holds each green until the maximum allowed.

We expect the proposed controller to produce the lowest delay in every scenario:

1) *Balanced demand.* The split between directions rises and falls in a recurring pattern that the Transformer learns from its training data. The proposed controller can therefore shift green toward the direction whose demand is about to increase, before its queue builds up. We expect this scenario to show the largest delay reduction relative to both baselines.
2) *Directional shift.* The sudden change lies outside the recurring pattern, so forecasts will be less accurate right after the shift. However, the controller re-plans every 5 s with updated queue estimates, so it should quickly move green toward the busier direction. The fixed-time plan cannot adapt at all, and the queue-responsive controller switches only after a large queue difference appears, so we still expect the proposed controller to have the lowest delay.
3) *Detector failure.* When detector reports stop arriving, the controller no longer knows how many vehicles are waiting, so both the proposed and queue-responsive controllers switch to the same predefined fixed timing plan. During the failure, all three controllers therefore follow the same signal rule. Before the failure and after reports resume, the proposed controller optimizes normally, so we expect its total delay to remain the lowest.
4) *Heavy demand.* More vehicles arrive than the intersection can serve, so every controller will end the simulation with vehicles still waiting. However, each phase change wastes about 4 s of yellow and all-red, during which almost no vehicles cross. Because the queues cannot clear, the proposed controller keeps each green as long as allowed and changes phases less often. It therefore wastes less time and serves more vehicles, so we expect its delay to remain lower than the baselines.

How we will judge the predictions. We run every controller on the same simulated traffic. For each run, we subtract each baseline's delay from the proposed controller's delay, so a negative number means our controller saved delay. Our expectations are confirmed if this difference is negative in almost every run of each scenario, with the largest savings under balanced demand. To confirm that the savings come from forecasting, we will also run the same optimizer without the Transformer forecast. Under balanced demand and after the directional shift, removing the forecast should increase delay. If it does not, the advantage of our controller would come from the optimization alone rather than from forecasting, and we would revise our explanation.

B. Minimum Simulation Design

Our simulation is a bounded isolated intersection (Fig. 1) that contains the elements of the minimum design: arrivals, exits, a bottleneck, and a controlled entry queue.

Bounded road and arrivals. The intersection has four approaches with two through lanes each. Each approach is 200 m long, so it stores up to 30 stopped vehicles per lane (Sec. I-B). Vehicles arrive at the upstream end of each lane as a random (Poisson) process whose rate varies over time. Under balanced demand, the north–south and east–west rates oscillate in opposite directions with a 300-s period, so total demand stays constant while the split between directions changes. Turns, lane changes, and pedestrians are excluded.

Bottleneck and exits. The signalized stop line is the bottleneck: each lane can discharge only during its green, at the rate produced by the IDM (Sec. I-C). Phase 1 serves north- and southbound traffic, and Phase 2 serves east- and westbound traffic. All controllers use 10–50 s of green, a 3-s yellow, and a 1-s all-red that is extended until the intersection is clear.

Entry queue. When an arriving vehicle finds no space at the upstream end of its lane, it waits in an entry queue outside the modeled road and enters only when space becomes available. These vehicles are recorded as unmet demand rather than discarded, and their waiting counts as delay. A vehicle's delay is the extra time it needs to reach the stop line compared with driving freely at v_0, including time in the entry queue.

Expected baseline response. Under fixed timing with demand below capacity, the queue should form during red and fully clear during green. Once demand exceeds the limit of Sec. I-A, the residual queue should grow each cycle until its tail reaches the 200-m boundary, after which further growth appears in the entry queue. Queue-length traces, cumulative curves, and entry-queue counts will show whether this happens.

C. Sensing, Forecasting, and Control

Detection and estimation. Each approach has loop detectors at its upstream end and at the stop line, reporting counts every 10 s. Following Sec. I-B, the number of vehicles on an approach is estimated as the difference between cumulative arrival and departure counts. When no valid report has arrived within the last 10 s, the controllers apply a predefined fallback timing plan.

Transformer forecaster. At each prediction time t, the model takes the historical sequence of the last $L = 12$ ten-second arrival counts of all approaches, $\mathbf{X}_t = [\boldsymbol{\lambda}_{t-L+1}, \ldots, \boldsymbol{\lambda}_t]$, augmented with time-of-day and day-of-week features $\mathbf{C}_t$, and predicts

$$\hat{\lambda}_{i,t+1} = f_\theta(\mathbf{X}_t, \mathbf{C}_t)_i, \qquad i \in \mathcal{A}. \tag{10}$$

To obtain forecasts over the horizon, each prediction is fed back into the input sequence, removing the oldest observation; six iterations give a 60-s forecast. The model f_θ is a small encoder (one layer, four attention heads) trained on 30 synthetic days that are never used for evaluation.

Controllers. We compare three controllers under identical signal constraints. (1) *Fixed-time control* gives each phase 30 s of green, chosen for the representative balanced demand. (2) *Queue-responsive control* switches when the opposing phase has more than two vehicles more than the current phase. (3) *Forecast-informed optimization* (ours) uses the Transformer forecasts in (7)–(8). Every 5 s, it evaluates hold-or-switch sequences over $H = 60$ s, keeps the 24 best partial sequences (beam search), implements the first decision, updates its observations, and repeats. As proposed in our original plan, an ablation runs the same optimizer without the forecast, planning from the current queue estimates only, to isolate the contribution of the Transformer.

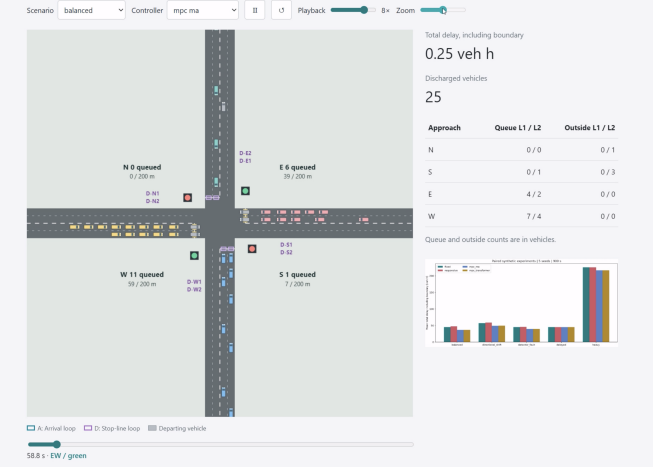


Fig. 2. Initial simulation environment and results.

D. Feasibility and Next Steps

As illustrated in Fig. 2, the intersection simulation, detectors, the three controllers, and result

TABLE II

SIMULATION SCENARIOS

Scenario	Implementation
Balanced demand	comparable average demand on all approaches; direction split oscillates with a 300-s period
Directional shift	from 450 s, north/southbound demand rises and east/westbound demand falls
Detector failure	reports missing from 360 to 585 s, or delivered 30 s late
Heavy demand	balanced profile scaled by 2.1

checks already run end to end for the four scenarios in Table II. Each scenario is simulated for 900 s, with identical demand realizations, initial queues, and simulation horizons across controllers. The main risks are the simplified queue model inside the optimizer and the synthetic demand. Next, we will:

1) measure saturation flow from simulated queue discharge, compare it with (6), and use it in the optimizer;
2) justify the fixed-time plan with Webster's cycle;
3) implement the ablation without the forecast, and a sensitivity study relating the Transformer's forecast error and the observation delay to the resulting delay;
4) repeat all experiments with more random seeds and evaluate total and approach-specific delay, throughput, maximum and residual queue lengths, unmet demand, and service imbalance between approaches, using summary statistics, 95% confidence intervals, and signal phase timelines with the corresponding queue evolution, reporting every scenario including those without improvement;

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APPENDIX

Components. Younggun Kim: Transformer forecasting model, simulation development, evaluation, and report drafting. Manuel A. Martinez Garcia: conceptualization, rolling-horizon optimizer, evaluation, and report drafting and finalization. We jointly designed the experiments, reviewed the full text, and are each responsible for the whole submission.

AI-use disclosure. Claude generated the initial simulation code, documentation, and pilot synthetic experiments. Claude was used to check our earlier draft against the assignment and to restructure and edit this text from our manuscript and code. We checked the equations, parameter values, and citations against the code and sources, and we take responsibility for the content.